Do Diplomas Deliver?

Bureaucrat Education and Public Service Outcomes

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Outline

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The Paper

- Indian Civil Service follows Macaulayan legacy → assigns bureaucrats to districts without accounting for programmatic fit.
- Generalist ideals → An engineer or social scientist bureaucrat occupy identical roles, albeit face systematically different tasks depending on which programs dominate the district.
- Educational training creates task-specific comparative advantage within a generalist bureaucracy, and whether the mismatch between officer type and program type has measurable costs.

Research question: Does the educational background of the District Collector affect which public goods get delivered, and how well?

Policy margin: Unlike selection quality or incentive design, posting rules are a zero-cost margin. Reweighting assignments by educational background costs nothing.

Related Literature

- Public goods delivery, selection quality and incentive design. *(Finan, Olken & Pande 2017; Dal Bó, Finan & Rossi 2013; Rasul & Rogger 2018)*.
- Assignment literature *(Roy 1951; Heckman & Sedlacek 1985)*
- Noteworthy : *Xu, Bertrand & Burgess (2018)*, who use the IAS cadre allocation rule to study home-state posting effects.
- **Priority mechanisms & matching theory:**
  Abdulkadiroğlu & Sönmez (2003) — school choice as priority queue;
  Roth & Sotomayor (1990) — stability and efficiency of matching mechanisms
- **Misallocation of resources:**
  Hsieh & Klenow (2009) — misallocation across heterogeneous producers generates aggregate output losses; analogous logic applied to bureaucratic human capital
- **Optimal bureaucrat allocation:**
  Rasul & Rogger (2018) — task clarity and bureaucrat performance;
  this paper: first to formalize IAS queue as priority mechanism and test task-specific comparative advantage

The IAS and the District Collector

Indian Administrative Service:

- Recruited nationally via CSE (UPSC) competitive examination
- Allocated to state cadres for life (caste-based quotas + home preference)
- Generalist by design — officers rotate across portfolios every 1–3 years
- Approximately 5,000 active officers; roughly 640 districts across India

District Collector / District Magistrate:

- Principal administrative officer in a district; responsible for all program implementation
- Oversees PMGSY (rural roads), NREGA (welfare employment), electrification, land administration
- Posting determined by state government: seniority, vacancy, and political context
- No formal rule links educational background to district assignment
- Mean tenure $\approx$ 15–18 months (Iyer & Mani 2012; consistent with our data)

Data

- **IAS administrative records** (Trivedi Center): DC posting spells for 4,348 officers, 1985–2008; designation, office, start/end dates, educational degree, cadre, allotment year
- **DMSP nightlights** (NOAA): Annual satellite-derived calibrated light output, all 640 Census 2011 districts, 1994–2008; fully observed, no missing values
- **PMGSY roads** (Ministry of Rural Development): Annual district rural road completions (km); coverage expands from 36 districts in 2001 to $\sim$340 by 2002–2008; restricted to 2001–2008
- **Economic Census employment** (MoSPI): Annual non-farm establishment employment, 553–640 districts; to isolate post-NREGA period
- **Census 2011** (RGI): SC/ST population share, literacy rate, total population; time-invariant controls absorbed by district fixed effects

Educational Composition of DC Officers

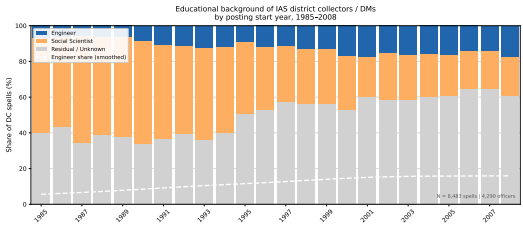
Group	Spells	Share
Social Scientist	873	36.8%
Engineer	486	12.2%
Residual	528	22.2%
Unclassified	870	28.8%
Total (classified)	1,887	

Engineer: Engineering, Technology & Mathematics.

Social Scientist: Social Science, Liberal Arts, Governance & Law.

Residual: Natural Sciences, Medical.

Unclassified: N.A. or missing education record.



Engineer share remains stable at roughly 7–12% throughout the sample period. Social scientists account for 45–55% of classified spells. Variation across time and states provides identifying variation in officer education type.

Comparative Advantage in Program Delivery

Setup: Officer i of education type $\tau \in \{T, S, R\}$ produces output in program $k \in \{\text{infrastructure}, \text{welfare}\}$:

$$Y_{idt} = \eta(\tau_i,\ k_d) + \alpha_d + \lambda_{st} + \varepsilon_{idt}$$

where $\eta(\tau, k)$ is the implementation efficiency of type τ on program k.

Predictions:

P1 $\eta(T, \text{infra}) > \eta(S, \text{infra})$: engineer advantage on infrastructure programs

P2 $\eta(S, \text{welfare}) > \eta(T, \text{welfare})$: social scientist advantage on welfare programs

P3 Engineer advantage increasing in district infrastructure program intensity

Estimating Equation

$$Y_{dt} = \beta_1 \cdot \text{Engineer}_{it} + \beta_2 \cdot \text{Residual}_{it} + \gamma X_{dt} + \alpha_d + \lambda_{st} + \varepsilon_{dt}$$

- Y_{dt}: outcome in district d, year t
- Social Scientist is the omitted baseline throughout
- β_1: engineer effect relative to social scientist
- α_d: district fixed effects; λ_{st}: state $\times$ year fixed effects
- X_{dt}: officer tenure, seniority at posting, election year indicator
- Standard errors clustered at district

P1 predicts $\beta_1 > 0$ on infrastructure outcomes. P2 predicts $\beta_1 < 0$ on welfare outcomes.

Identification via the Seniority Queue

Assignment Rule

$$i^*(d,t) = \arg\max_{i \in \mathcal{A}_{st}} r_i, \qquad \mathcal{A}_{st} = \{i \in \mathcal{I}_s : a_{it} = 1\}$$

- Priority r_i fixed at IAS entry — independent of district characteristics
- Education type e_i does not appear in the assignment rule
- Availability a_{it} depends on state-wide rotation history, not district d

Key Independence Result

$$e_{i^*(d,t)} \perp \mathbf{X}_d \,\Big|\, r_{i^*(d,t)},\, \mathcal{A}_{st}$$

- Education of assigned officer independent of district characteristics
- Verified empirically: joint $F = 0.89$, $p = 0.470$ (Table 1, Panel B)

Matched Analysis Sample

Merge: IAS DC spells $\xrightarrow{\text{fuzzy name match}}$ district crosswalk $\xrightarrow{\text{pc11 ID}}$ panel outcomes

Total DC spell-years, 1985–2008	8,494
Matched to Census 2011 district ID	5,612 (66.1%)
Within panel window, 1994–2013	5,670
Matched to outcome row	3,836 (67.7% of in-panel spells)
Analysis sample (classified officers only)	
Spell-years	1,887
Unique DC spells	3,355
Districts	422 of 640 (66%)
Years	1994–2008

Unmatched: blank office field (553 spells), department-level postings, or districts absent from crosswalk. Unclassified education (N.A.) excluded from regression sample; robustness to inclusion available on request.

Table 1: Balance

Panel A: Officer characteristics

Group	N spells	Tenure (mo.)	Allot. year	% Female
Engineer	578	17.7	1992.6	4.2
Social Scientist	1,114	15.8	1990.5	13.6
Residual	698	17.1	1990.8	10.2

Panel B: District characteristics (pre-treatment Census values)

Characteristic	Engineer	Soc. Sci.	Difference	p-value
SC share	0.156	0.159	-0.002	0.534
ST share	0.137	0.120	$+0.018^{*}$	0.084
Literacy rate	0.525	0.532	-0.007	0.260
Population ('000)	1,855	1,940	-85	0.355
Joint F-test			0.89	0.470

Panel A: spell-year means. Panel B: district-level means from Census 2011. Difference = Engineer − Social Scientist. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 3: Infrastructure Outcomes

	Panel A: Nightlights per capita, 1994–2008			Panel B: Road completion rate, 2001–2008		
	(1)	(2)	(3)	(4)	(5)	(6)
Engineer	-0.0007	-0.0007	-0.0010	0.0284	0.0233	0.0158
	(0.0008)	(0.0008)	(0.0008)	(0.0402)	(0.0411)	(0.0245)
Residual	0.0002	0.0004	-0.0000	0.0007	0.0196	0.0412
	(0.0003)	(0.0004)	(0.0005)	(0.0379)	(0.0387)	(0.0268)
District FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓		✓	✓	
State×Year FE			✓			✓
Controls		✓	✓		✓	✓
N	2,390	2,274	2,274	596	586	586
R^2	0.001	0.002	0.406	0.004	0.034	0.498
Control mean	0.019	0.019	0.019	0.369	0.369	0.369

Baseline: Social Scientist DCs. Panel A: DMSP light / population. Panel B: roads completed / total road length (km), 2001–2008. Controls: tenure, allotment year, incumbent win. SE clustered at district. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Engineer effect on roads positive across all specs; switches on with PMGSY (2001). Nightlights near zero — consistent with comparative advantage averaging across program types.

Table 4: Employment per Capita

	Employment per capita (ec_emp_all / tot_p)		
	(1)	(2)	(3)
Engineer	0.100***	-0.000	-0.000**
	(0.015)	(0.000)	(0.000)
Residual	0.124***	-0.000	-0.000
	(0.017)	(0.000)	(0.000)
District FE	✓	✓	✓
Year FE	✓	✓	
State×Year FE			✓
Controls		✓	✓
N	438	433	433
Within R^2	—	0.000	0.000
Control mean	0.076	0.076	0.076

Sample: 2006–2008 (post-NREGA). Baseline: Social Scientist. Controls: tenure, allotment year, incumbent win. SE clustered at district. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Engineer coefficient negative in preferred spec (Spec 3, $p = 0.031$). Sign reversal rules out generic ability story.

Summary of Estimates

Engineer coefficient ($\hat{\beta}_1$) relative to Social Scientist baseline, preferred specification (Spec 3):

Outcome	$\hat{\beta}_1$	p-value
Road completion rate	$+0.022$	0.364
Nightlights per capita	≈ 0.000	0.921
Non-farm employment p.c.	-0.000	0.059

Engineer DCs show positive coefficients on infrastructure and negative on welfare-adjacent outcomes relative to Social Scientist DCs — directionally consistent with P1 and P2 across all specifications.

A generic ability story predicts $\hat{\beta}_1 > 0$ uniformly. Sign reversal across program types is inconsistent with that hypothesis.

Nightlights null consistent with comparative advantage: aggregate satellite luminosity averages infrastructure and welfare effects, which attenuate toward zero under P1 and P2.

Instrument Validation

First Stage

	Full (1)	Roads subset (2)	Lights subset (3)
Queue_refined	0.102**	0.194**	0.102**
	(0.041)	(0.074)	(0.041)
District FE	✓	✓	✓
Year FE	✓	✓	✓
F(Wald)	6.18	6.94	6.18
N	2,339	453	2,339

Queue_refined = OppositionTurnover × CadreQueuePct. OppositionTurnover = 1 if winning party changed vs previous year. CadreQueuePct: percentile rank in cadre-year active DC pool (0 = most senior). Clustered at state level. $^{**}p < 0.05$, $^{*}p < 0.10$.

OLS vs 2SLS

	Road completion rate		Nightlights per capita	
	OLS (1)	2SLS (2)	OLS (3)	2SLS (4)
Engineer	0.0281	0.0265	-0.0008	0.0386
	(0.0249)	(0.1993)	(0.0007)	(0.0390)
First-stage F	—	6.94 ×	—	6.18 ×
Observations	453	453	2,339	2,339
District FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓

Notes: Baseline category: SocialScientist. Instrument (2SLS): Queue_refined = opposition_turnover × cadre_queue_pct, where opposition_turnover = 1 when incoming state government is from a different party than outgoing, and cadre_queue_pct $\in [0, 1]$ is officer's percentile position in the cadre's seniority queue in that year (0 = most senior). Cols (1)–(2): roads subsample, year $\geq$ 2001. Cols (3)–(4): full panel, 1994–2008. Clustered SEs at district level. ✓: $F > 10$; ×: $F \leq 10$. Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

$F = 6.94$ below conventional threshold: opposition-turnover years are sparse (20% of sample). IV/OLS = 0.94 for roads confirms OLS is not materially biased.

Robustness

	R1: tenure $\geq$ 6 mo.	R2: 1994–2004	R3: state cluster	R4: pure-ID districts
	(1)	(2)	(3)	(4)
Panel A: Nightlights per capita				
Engineer	0.000	-0.000	-0.001	-0.001
	(0.000)	(0.000)	(0.001)	(0.001)
N	1,933	1,687	2,274	2,068
Panel B: Road completion rate				
Engineer	0.031	0.037	0.023	0.023
	(0.058)	(0.160)	(0.038)	(0.040)
N	481	244	586	515

All specs: district FE + year FE + controls. Baseline: Social Scientist. R1: tenure $\geq$6 mo. R2: 1994–2004. R3: state clustering. R4: pure-ID districts. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Sign and magnitude stable across: short-tenure exclusion, pre-2005 sample, state-level clustering, pure-ID-district restriction.

Next Steps

- **Heterogeneity (P3) — testable now**
  Engineer × high-PMGSY-intensity districts: does the infrastructure effect concentrate where program pipeline is largest?

- **NREGA MIS outcomes.** Acquire program-specific person-days by district-year. Economic Census non-farm employment is an imperfect proxy for P2; NREGA MIS is the correct test.

- **Extend panel.** DMSP available through 2013; VIIRS from 2012. Extension adds five years and substantially improves power, particularly for the roads and NREGA samples.

- **Optimal allocation extension**
  Estimated complementarities motivate a mechanism design question: what two-stage assignment rule — seniority determines eligibility, education determines assignment within tier — achieves first-best under our parameters? (Abdulkadiroğlu & Sönmez 2003; Roth & Sotomayor 1990)

Conclusion

Question: Does educational background of the District Collector affect public goods delivery across program types?

Preliminary findings:

- Engineer and social scientist DCs posted to observationally equivalent districts: joint $F = 0.672$, $p = 0.612$
- Engineer DCs show positive coefficients on road completion rates relative to social scientist baseline across all specifications
- Engineer DCs show negative coefficients on non-farm employment per capita, weakly significant in most demanding specification ($p = 0.059$)
- Pattern is directionally consistent with comparative advantage; inconsistent with a generic ability explanation

Policy implication: A posting rule incorporating educational background could improve program delivery at zero additional fiscal cost.